

Red wine may hold the key to next-generation wearable technology

A team of scientists from University of Manchester is seeking to kick-start a wearable technology revolution by creating flexible fibres and adding acids from red wine. Extracting tannic acid from red wine, coffee or black tea, led the team of scientists to develop much more durable and flexible wearable devices. Addition of tannins improved the mechanical properties of materials such as cotton to develop wearable sensors for rehabilitation monitoring, drastically increasing the devices' lifespan.

The team has developed wearable devices such as capacitive breath sensors and artificial hands for extreme conditions by improving the durability of flexible sensors.

Previously, wearable technology has been subject to failure after repeated bending and folding, which can interrupt the conductivity of such devices due to tiny micro cracks. Improv-

ing this could open the door to more long-lasting integrated technology.

Dr Xuqing Liu, who led the research team, commented, "We are using this method to develop new flexible, breathable, wearable devices. The main research objective of our group is to develop comfortable wearable devices for flexible human-machine interface.

"Traditional conductive material suffers from weak bonding to fibres, which can result in low conductivity. When red wine, coffee or black tea is spilled on a dress, it is difficult to get rid of these stains. The main reason is that they all contain tannic acid, which can firmly absorb the material on the surface of the fibre. This good adhesion is exactly what we need for durable wearable, conductive devices."

New e-skin gives robots and prosthetics an exceptional sense of touch

Robots and prosthetic devices may soon have a sense of touch equivalent to, or better than, human skin with Asynchronous Coded Electronic Skin (ACES), an artificial nervous system developed by a team of researchers at National University of Singapore (NUS).

The new electronic skin, or e-skin, system has achieved ultra-high responsiveness and robustness to damage, and can be paired with any kind of sensor skin layers to function effectively as e-skin.

Drawing inspiration from the human sensory nervous system, the team spent a year and a half developing a sensor system that could potentially perform better. While the ACES electronic nervous system detects signals like the human sensor nervous system, it is made up of a network of sensors

connected via a single electrical conductor, unlike the nerve bundles in the human skin. It is also unlike existing e-skins, which have interlinked wiring systems that can make these sensitive to damage and difficult to scale up.

ACES can detect touches more than a thousand times faster than the human sensory nervous system. For example, it is capable of differentiating physical contacts between different sensors in less than sixty nanoseconds—the fastest ever achieved for an e-skin technology—even with a large numbers of sensors. ACES-enabled skin can also accurately identify the shape, texture and hardness of objects within ten milliseconds, which is ten times faster than the blinking of an eye. This is enabled by high fidelity and capture speed of the ACES system.

Carnegie Mellon and Facebook AI beats professionals in six-player poker

An artificial intelligence (AI) program developed by Carnegie Mellon University in collaboration with Facebook AI has defeated leading professionals in six-player No-Limit Texas Hold'em poker, the world's most popular form of poker.

The AI, called Pluribus, defeated poker professional Darren Elias, who holds the record for most World Poker Tour titles; and Chris 'Jesus' Ferguson, winner of six World Series of Poker events. Each professional separately played 5000 hands of poker against five copies of Pluribus.

In another experiment involving thirteen professionals, all of whom have won more than one million dollars playing poker, Pluribus played against

five of them at a time for a total of 10,000 hands and again emerged victorious.



Pluribus playing a six-player game (Credit: Facebook)